

Math 162

Problem set 7 solutions

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1. (a) Observe that

$$\frac{n!}{n^n} = \frac{1}{n} \cdot \frac{2}{n} \cdots \frac{n-1}{n} \cdot \frac{n}{n} \leq \frac{1}{n},$$

since $\frac{k}{n} \leq 1$ for any $k \leq n$. Since $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$, it follows that $\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$ by the squeeze theorem.

(b) Since $\lim_{x \rightarrow \infty} \frac{1}{x} \log a = 0$, it follows from the continuity of exp that

$$\lim_{x \rightarrow \infty} a^{1/x} = \lim_{x \rightarrow \infty} \exp\left(\frac{1}{x} \log a\right) = \exp 0 = 1.$$

As shown in class, it follows that

$$\lim_{n \rightarrow \infty} \sqrt[n]{a} = \lim_{n \rightarrow \infty} a^{1/n} = \lim_{x \rightarrow \infty} a^{1/x} = 1.$$

(c) For this, recall that $\lim_{x \rightarrow \infty} \frac{\log x}{x} = 0$. The continuity of exp then implies that

$$\lim_{x \rightarrow \infty} x^{1/x} = \lim_{x \rightarrow \infty} \exp\left(\frac{1}{x} \log x\right) = \exp 0 = 1.$$

As in part (b), we deduce that

$$\lim_{n \rightarrow \infty} \sqrt[n]{n} = \lim_{n \rightarrow \infty} n^{1/n} = \lim_{x \rightarrow \infty} x^{1/x} = 1.$$

(d) Suppose without loss of generality that $a \geq b$ (otherwise interchange a and b). If $a = b$, then

$$\lim_{n \rightarrow \infty} \sqrt[n]{a^n + b^n} = \lim_{n \rightarrow \infty} \sqrt[n]{2a^n} = \lim_{n \rightarrow \infty} a \sqrt[n]{2} = a.$$

by part (b). If $a > b$, then in particular $a \neq 0$, so we have the factorization

$$\sqrt[n]{a^n + b^n} = \sqrt[n]{a^n \left(1 + \left(\frac{b}{a}\right)^n\right)} = a \sqrt[n]{1 + \left(\frac{b}{a}\right)^n}.$$

It therefore suffices to show that

$$\lim_{n \rightarrow \infty} \sqrt[n]{1 + \left(\frac{b}{a}\right)^n} = 1.$$

Let $c := \frac{b}{a} < 1$. By definition, for any $x > 0$ we have

$$(1 + c^x)^{1/x} = \exp\left(\frac{1}{x} \log(1 + c^x)\right).$$

Since $0 < c < 1$, we have $\lim_{x \rightarrow \infty} c^x = 0$, and hence

$$\lim_{x \rightarrow \infty} \log(1 + c^x) = \log 1 = 0$$

by the continuity of log. Thus

$$\lim_{x \rightarrow \infty} \frac{1}{x} \log(1 + c^x) = 0,$$

so

$$\lim_{x \rightarrow \infty} \exp\left(\frac{1}{x} \log(1 + c^x)\right) = 1$$

by the continuity of exp.

2. There are many ways to do this: here is one possible solution. For any $n \in \mathbb{N}$, we let $a_n := \frac{k}{n}$ where $0 < k < n$ is such that $|\frac{k}{n} - L|$ is minimal. Note that there may be two such k : in that case, pick one arbitrarily. It follows that $a_n - \frac{1}{2n} \leq L \leq a_n + \frac{1}{2n}$, i.e. $|a_n - L| \leq \frac{1}{2n}$ (if one of these bounds were violated, then L would be closer to $\frac{k-1}{n}$ or $\frac{k+1}{n}$ than $\frac{k}{n}$). Thus $\lim_{n \rightarrow \infty} a_n = L$ by the squeeze theorem.

3. Observe that

$$\lim_{n \rightarrow \infty} \frac{1/n}{1/n^{1+1/n}} = \lim_{n \rightarrow \infty} \frac{n^{1+1/n}}{n} = \lim_{n \rightarrow \infty} n^{1/n} = 1,$$

where the last equality used #1(c). Since the harmonic series diverges, the limit comparison test implies that $\sum_{n=1}^{\infty} \frac{1}{n^{1+1/n}}$ diverges.

4. Fix $\epsilon > 0$. Since $\lim_{n \rightarrow \infty} a_n = L$, there exists $N_0 \in \mathbb{N}$ such that $n > N_0$ implies that $|a_n - L| < \frac{\epsilon}{2}$. Since the sequence (a_n) converges it is in particular bounded, and hence the sequence $(a_n - L)$ is also bounded. Let $M > 0$ be such that $|a_n - L| \leq M$ for all $n \in \mathbb{N}$. Finally, choose $N \in \mathbb{N}$ such that $N > \max(N_0, \frac{2MN_0}{\epsilon})$. Let $n \in \mathbb{N}$ be such that $n > N$. Then we have

$$\begin{aligned} \left| \left(\frac{1}{n} \sum_{k=1}^n a_k \right) - L \right| &= \left| \frac{1}{n} \sum_{k=1}^n (a_k - L) \right| \\ &\leq \frac{1}{n} \sum_{k=1}^n |a_k - L| \\ &= \frac{1}{n} \sum_{k=1}^{N_0} |a_k - L| + \frac{1}{n} \sum_{k=N_0+1}^n |a_k - L|. \end{aligned}$$

On one hand, we have

$$\frac{1}{n} \sum_{k=1}^{N_0} |a_k - L| \leq \frac{1}{n} \cdot N_0 M < \frac{N_0 M}{N} < \frac{\epsilon}{2}.$$

On the other hand,

$$\frac{1}{n} \sum_{k=N_0+1}^n |a_k - L| < \frac{1}{n} \cdot (n - N_0) \frac{\epsilon}{2} < \frac{\epsilon}{2}.$$

Combining these estimates with the above, we obtain

$$\left| \left(\frac{1}{n} \sum_{k=1}^n a_k \right) - L \right| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

as needed.